BME577: Drug Delivery **Spring 2025**
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Project 3: Delivery of Small Molecule Drugs to Metastatic Tumors

1) *Determine intracellular and extracellular concentration distributions of the drug in the tumor, i.e. C_i (r) and C_e (r) , at t = 30, 60, and 120 seconds.*

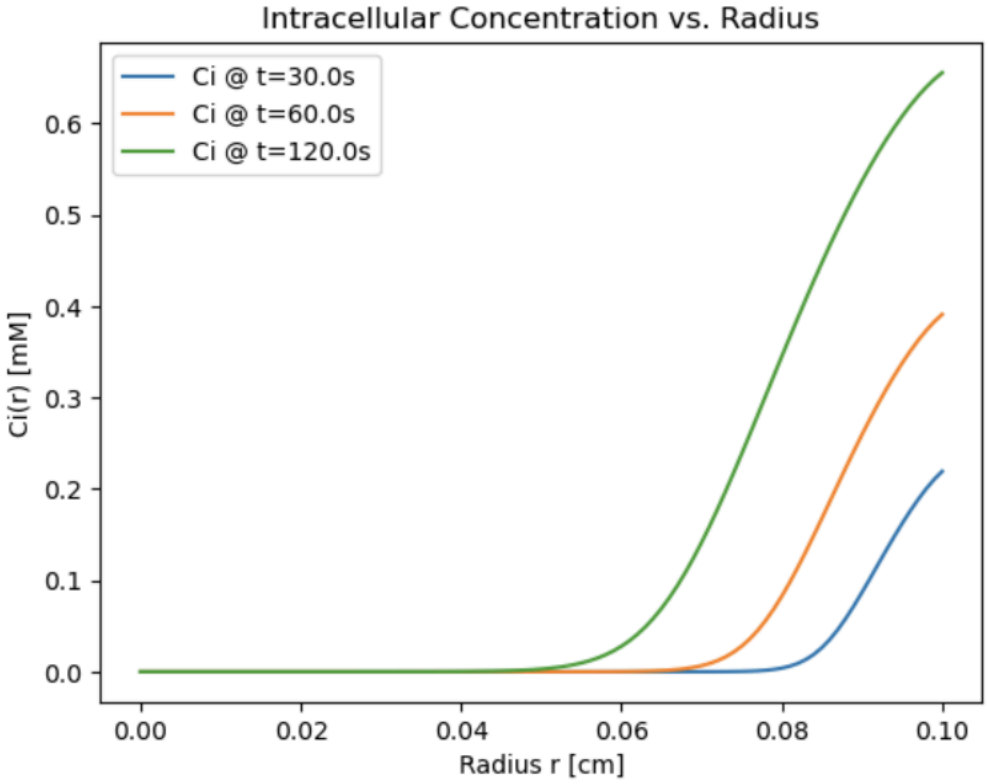


Figure 1. Intracellular concentration C_i at three time points (t = 30, 60, 120 s) in a spherical tumor of radius 0.1 cm. Each curve shows how C_i increases from the center (r = 0) toward the boundary (r = 0.1 cm) as time progresses.

Drug enters from the tumor boundary and diffuses inward. The cells near the boundary begin to accumulate drug first, leading to a steep radial gradient in intracellular concentration. Over time, the intracellular drug levels continue to rise and extend toward the tumor center, resulting in higher and broader C_i profiles at later times.

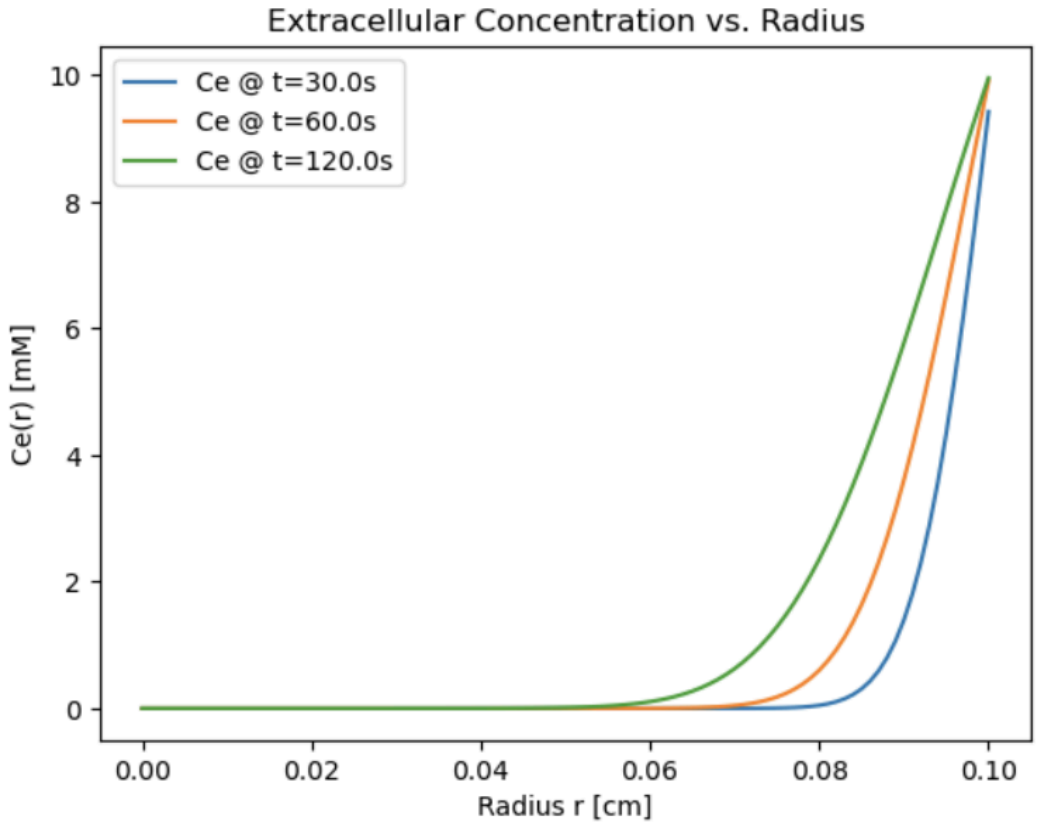


Figure 2. Extracellular concentration at three time points (t = 30, 60, 120 s) for a spherical tumor of radius 0.1 cm. Each curve shows how C_e builds up from the boundary (r = 0.1 cm) inward as time progresses.

Drug enters the tumor from the outer radius, causing a steep radial gradient in the early stages. Over time, this extracellular concentration front advances toward the center, raising C_e throughout the tumor. By 120 s, much of the tumor has been exposed to the drug, but the concentration still remains higher near the boundary where it is continuously replenished from the surrounding tissue.

2) *If V_{max} varies by a factor of 10, how does its increase or decrease affect C_i (r) and C_e (r) at t = 120 seconds?*

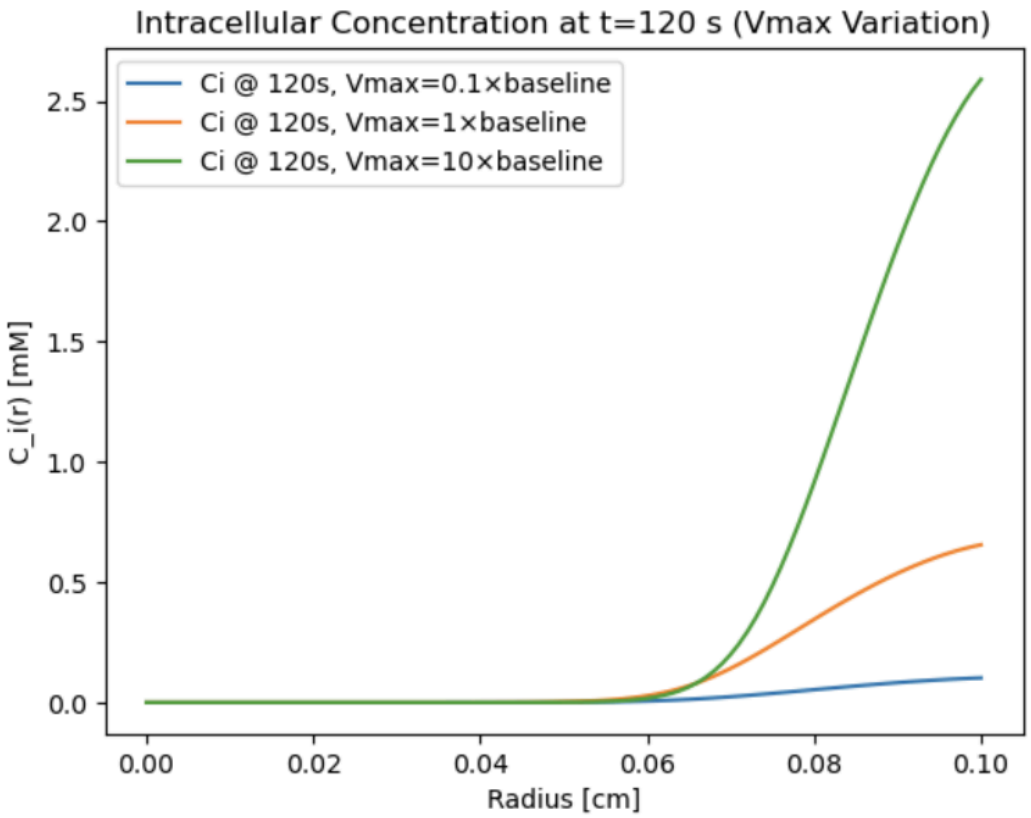


Figure 3. Intracellular drug concentration C_i at t = 120 s for three different values of V_{max} (0.1x baseline, 1x baseline, and 10x baseline). Each curve illustrates the radial distribution of the intracellular concentration across the tumor.

A higher V_{max} (10x baseline) greatly enhances the rate of drug uptake by cells, resulting in a larger intracellular concentration near the boundary where the extracellular concentration is highest. In contrast, lowering V_{max} significantly reduces intracellular accumulation, leaving more drug in the extracellular space and leading to a comparatively lower C_i across the tumor.

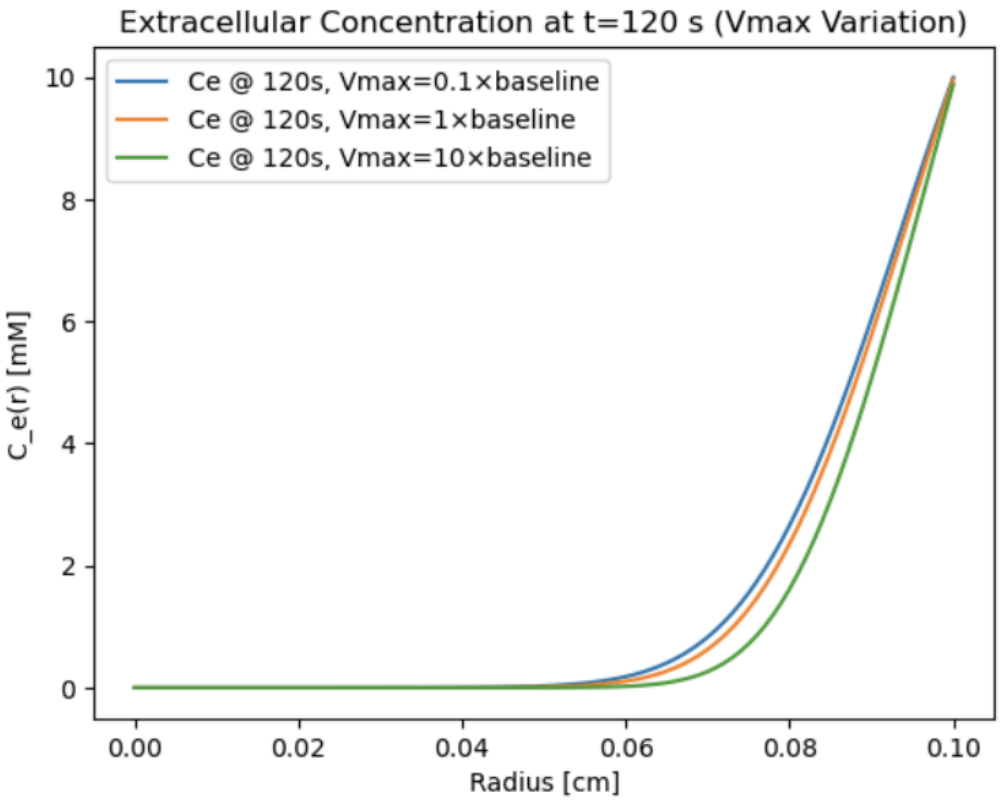


Figure 4. Extracellular drug concentration at t = 120 s for three different values of V_{max} (0.1x baseline, 1x baseline, and 10x baseline), plotted as a function of radial position in the tumor.

With higher V_{max}, cells remove more drug from the extracellular space, resulting in slightly lower C_e near the boundary where drug uptake is greatest. Conversely, a lower V_{max} leads to weaker uptake, so the extracellular concentration remains higher throughout the tumor at 120 s.

3) *Similarly, if K_m varies by a factor of 10, how does its increase or decrease affect C_i (r) and C_e (r) at t = 120 seconds?*

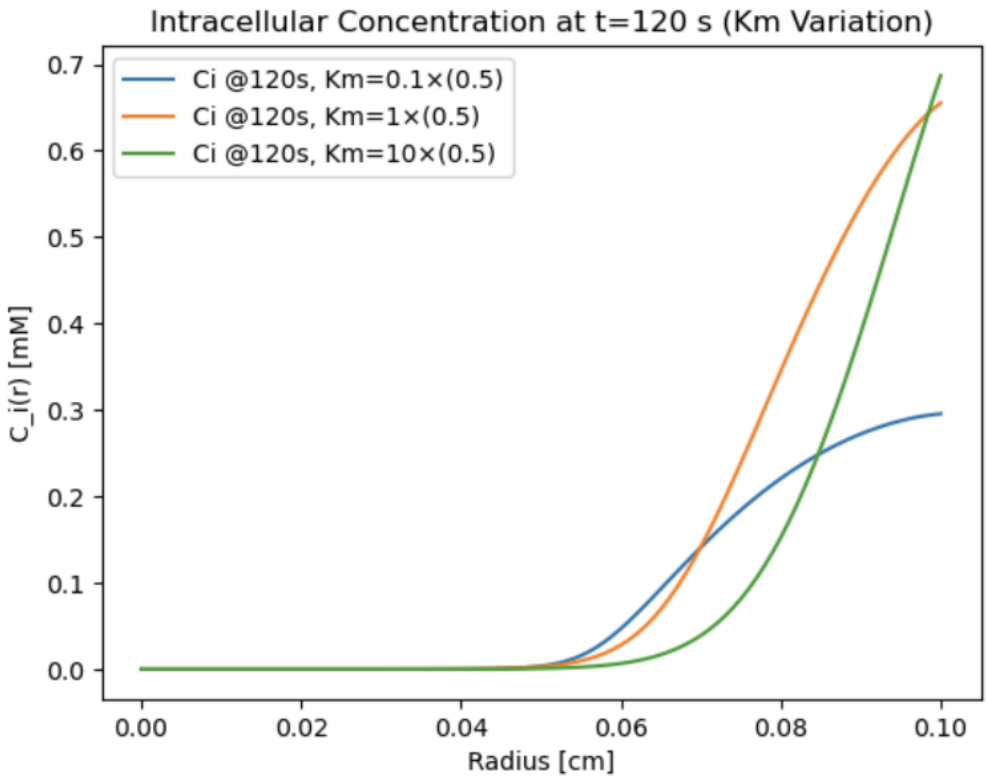


Figure 5. Intracellular drug concentration at t = 120 s for three different values of the Michaelis constant K_m (0.1x baseline, 1x baseline, and 10x baseline), plotted as a function of radial position in the tumor.

Decreasing K_m (the blue curve) makes the transporter closer to its saturating uptake rate at typical drug concentrations, leading to higher intracellular accumulation across the tumor. Increasing K_m (the green curve) raises the concentrations needed to reach saturation, lowering the net uptake rate under these conditions and thus reducing C_i. The baseline case (the orange curve) lies in between. Consequently, lowering K_m enhances intracellular drug levels, while raising it diminishes them, all else being equal.

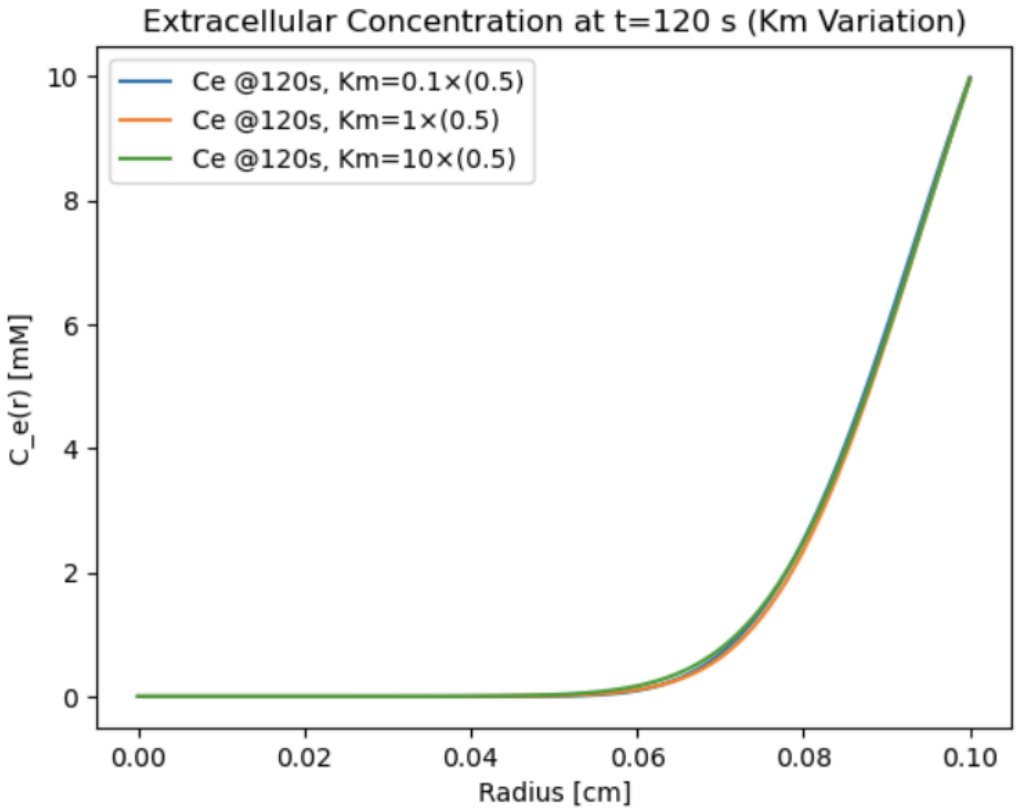


Figure 6. Extracellular drug concentration at t=120 s for three different values of the Michaelis constant K_m (0.1x baseline, 1x baseline, and 10x baseline), shown across the tumor radius.

When K_m is decreased, transporters reach their saturating uptake rate at lower concentrations, causing more rapid intracellular drug removal and slightly depleting the extracellular level—thus producing a curve that is marginally lower (especially near the boundary). When K_m is increased, the transporters are less efficient at moderate concentrations, so C_e remains marginally higher. Overall, changing K_m by an order of magnitude shifts the balance between cellular uptake and extracellular retention, but the overall shape of the C_e profile remains similar.

4) *Set all parameter values to their baseline levels, except for V_{max}, which should be 100,000 times smaller than its baseline level (assuming cellular uptake of the drug is negligible). Estimate the diffusion time required for the drug concentration at r = 0 (the center of the tumor) to reach 99% of the concentration in the surrounding tissue (reaching 9.9 mM). (Hint: run your computer code for different endpoint t values until C_i (0) is between 9.85 and 9.95 mM).*

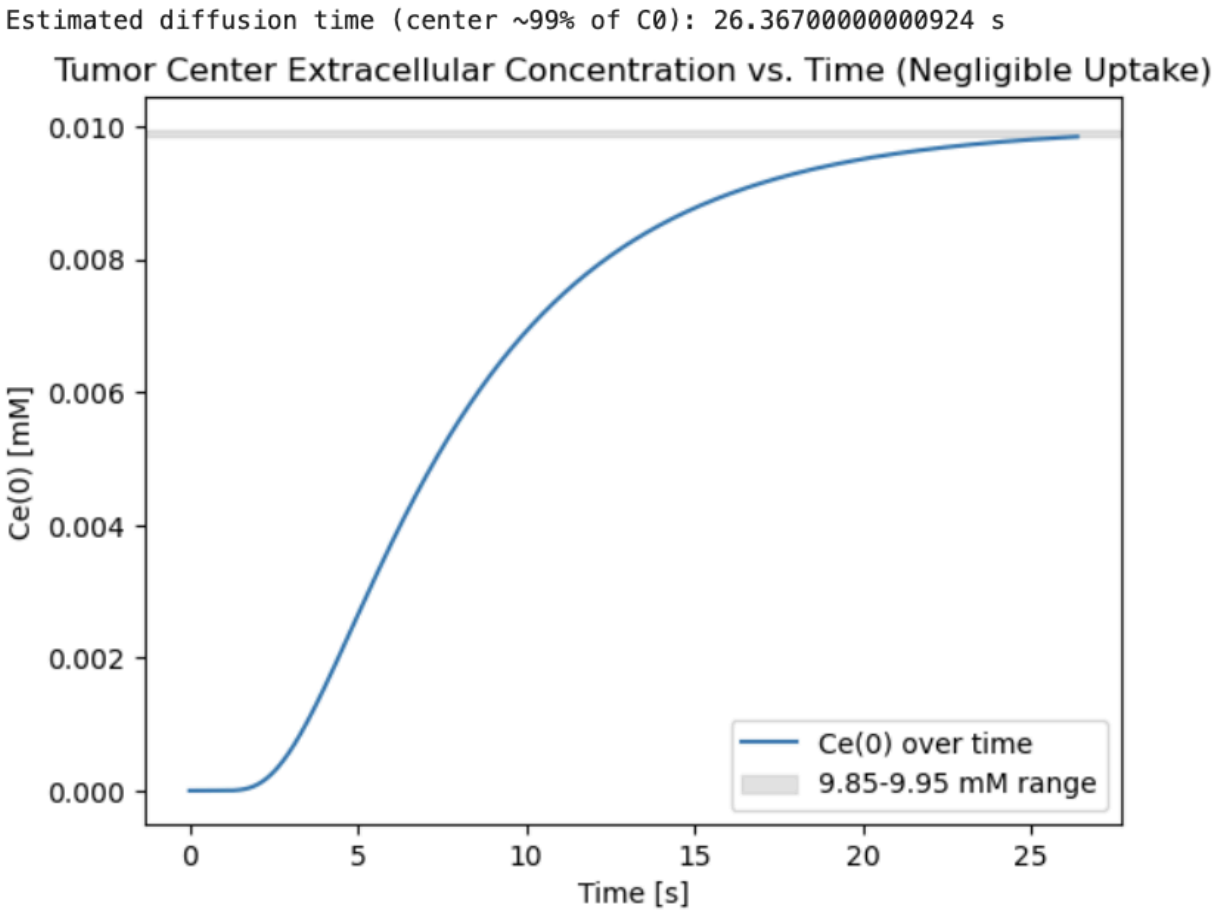


Figure 7. Time evolution of the extracellular drug concentration at the tumor center (r=0) under negligible cellular uptake. The horizontal shaded region (9.85–9.95 mM) indicates when the center reaches ~99% of the boundary concentration (~10 mM). The simulation shows that the tumor center achieves this concentration at approximately **26.37 s**, representing the characteristic diffusion time.

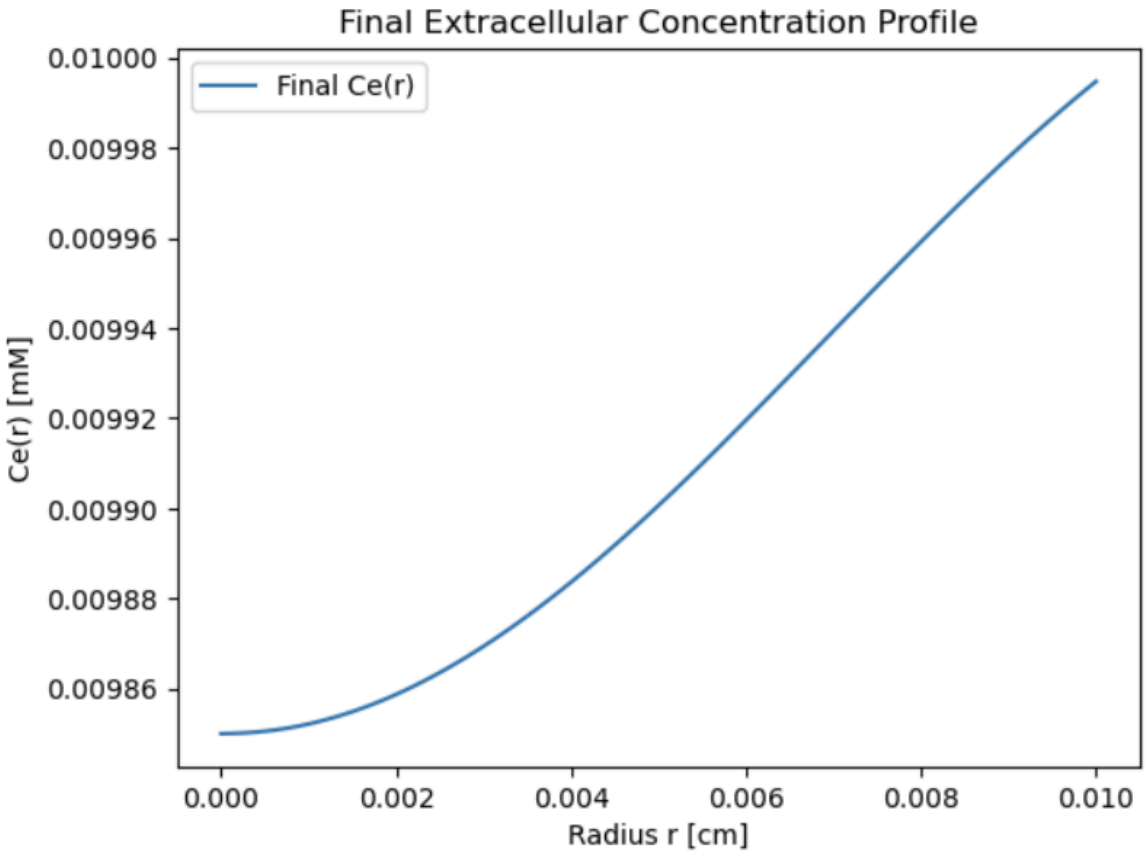


Figure 8. Final radial distribution of the extracellular concentration, C_e, at the end of the simulation when the tumor center has reached ~99% of the boundary concentration under negligible uptake. The near-uniform high-level concentration highlights that diffusive penetration is essentially complete, with only a slight gradient remaining between the center (r = 0) and the boundary (r = R).

In this scenario, cellular uptake is negligible (V_{max} reduced by 10^5), so the drug transport in the tumor is primarily governed by diffusion from the boundary at r = R. The time it takes for the tumor center (r = 0) to reach 99% of the external concentration (~9.9 mM) is therefore the characteristic diffusion time. Physically, once the boundary begins supplying drug at C_0, it must diffuse inward through the tumor tissue. Thus, a rough analytic estimate for purely diffusive transport often comes from t_{diff} approximately relating to R^2/D. However, exact times can differ depending on additional boundary flux conditions (here a Robin-type boundary with coefficient *k*) and other parameter details. Running the simulation numerically until C_e sits between 9.85 mM and 9.95 mM provides a direct measure of how quickly the drug concentration equilibrates to near C_0 at the tumor center.

5) *Repeat Exercise 4 for R = 500 um and 1000 um respectively, and compare the three diffusion time periods in your simulations with the estimation using $R^2/(2D)$ to verify that the diffusion time scales with R^2. (Hint: Compare the two plots of the diffusion time versus R. One of the c*urves is based on your computer simulation, while the other is based on $R^2/(2D)$.)

```
R = 100 μm => R_cm = 0.01 cm
  Simulation time: 26.367 s
  Theory time    : 25.000 s

R = 500 μm => R_cm = 0.05 cm
  Simulation time: 642.176 s
  Theory time    : 625.000 s

R = 1000 μm => R_cm = 0.1 cm
  Simulation time: 1000.000 s
  Theory time    : 2500.000 s
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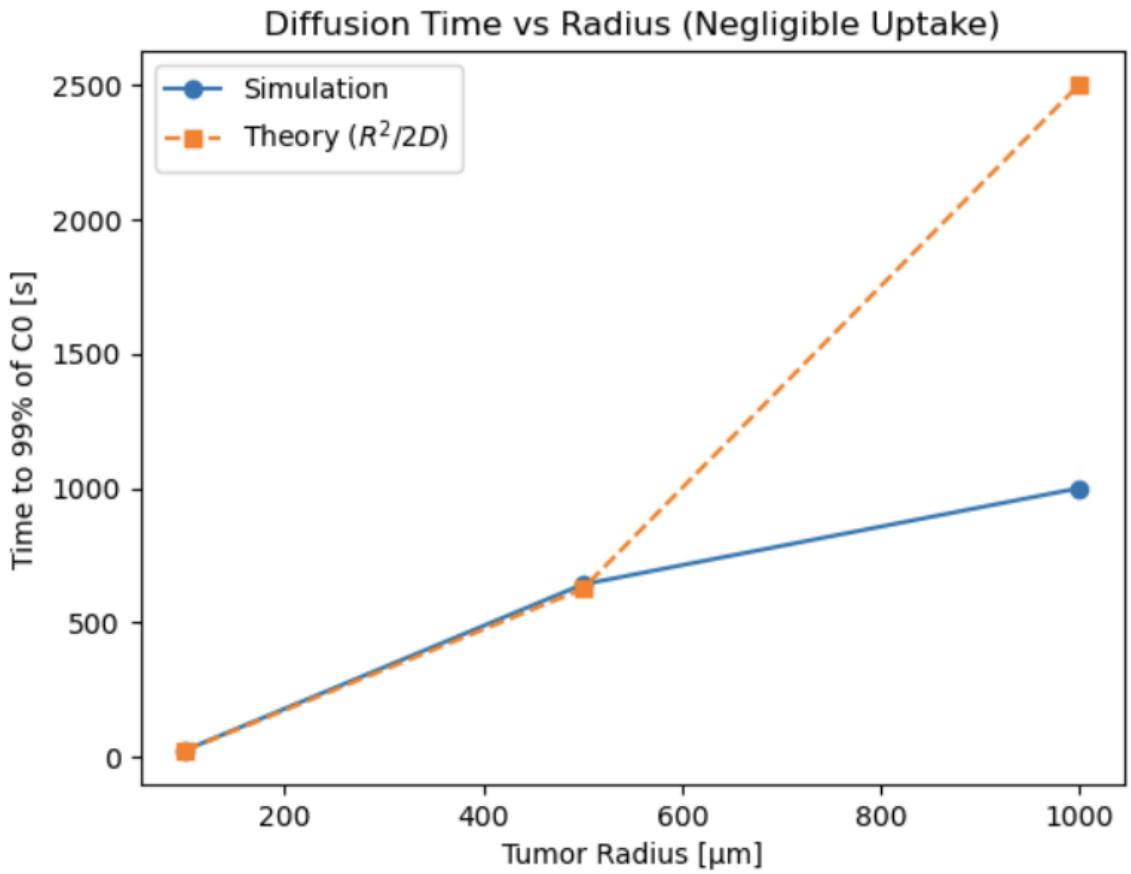


Figure 9. Comparison of the numerically computed diffusion times (blue) and the theoretical prediction $R^2/(2D)$ (orange) for three tumor radii (100 µm, 500 µm, and 1000 µm). The plot illustrates how the time required for the center to reach 99% of the boundary concentration scales approximately with R^2.

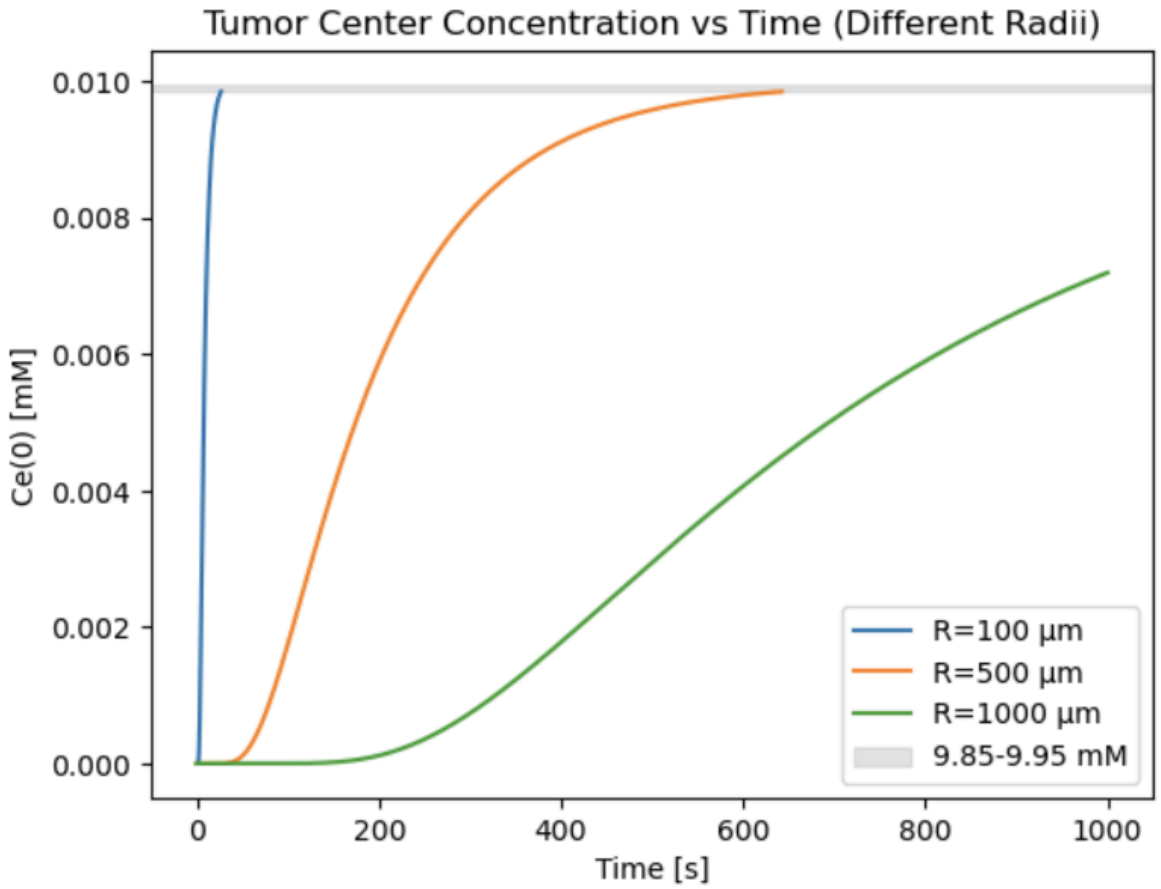


Figure 10. Time evolution of the tumor-center extracellular concentration for three different tumor radii (100 µm, 500 µm, 1000 µm) under negligible uptake. The shaded band at 9.85–9.95 mM represents ~99% of the boundary concentration (10 mM). As the radius increases, the center concentration takes longer to approach this near-equilibrium level, illustrating the strong dependence of diffusion time on tumor size.

Although there are minor differences due to the Robin-type boundary, the results confirm that the time to reach near-equilibrium scales roughly with R^2, meaning larger tumors take significantly longer for the drug to diffuse from the boundary to the center, as predicted by the simple quadratic diffusion-time formula.